And also the proportion of males (M) and Females (F) among the literates. Which one of the following statements can be said to be true beyond doubt?

(a) In 1970 half of the illiterates were women (Females).
(b) The proportion of literate males to the total population of males remained the same over the years.
(c) Males literacy did not improve over this period
(d) The ratio of female literates to male literates improved significantly over this period.

Ans. (c): According to above pie chart very concisely. Analyses that male literacy did not over this period. Hence option (c) is correct answer.

132.Six roads lead to a country. They may be indicated by letters X, Y, Z and digits 1, 2, 3. When there is storm, Y is blocked. When there are floods X, 1 and 2 will be affected. When road 1 is block, Z also is blocked. At a time when there are floods and a storm also blows, which road (s) can be used?

(a) Z and 2 (b) only Z
(c) only 3 (d) only Y

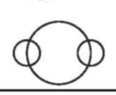
Ans. (c): Y is blocked for storms. For flood X, 1 and 2 are blocked. Z is blocked whenever 1 is blocked. So, for floods and storms, only unblocked road left is 3.

133.A truck, a car and a motor cycle have equal kinetic energies. If equal, stopping forces are applied and they stop after travelling a distance of X, Y and Z respectively, then:

(a) $X > Y > Z$ (b) $X < Y < Z$
(c) $X = Y = Z$ (d) $X \simeq 4Y \simeq 8Z$

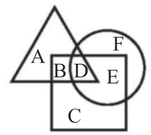
Ans. (c): According to the question the kinetic energy of all three vehicles is equal and stopping force is also equal, therefore X = Y = Z because work done by stopping force = loss of kinetic energy.

134.Which one of the following venn diagrams correctly illustrates the relationship among the classes: carrot, food vegetable?

(a)  (b) 
(c)  (d) 

Ans. (a): Both carrot and vegetables come under food category, while carrot comes under vegetable category.

135.In the given figure, the triangle represents girls the square represents sports persons and the circle represents coaches. The portion in the figure which represents girls who are sports persons but not coaches is the one labeled as



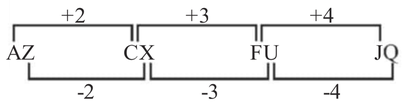
(a) A (b) B
(c) E (d) E

Ans. (b): It is the common area between the triangle and square but excluding the circle i.e. B.

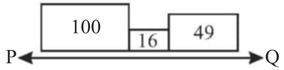
136.The next pair of letters in the series AZ, CX, FU _____ is

(a) JQ (b) KP
(c) IR (d) IV

Ans. (a): First letter in each pair increase by 2, 3, 4. While the second letter decreases by 2, 3, 4_____ respectively. So 4 letters after F is J and before U is Q. Hence, letter is JQ.



137.The following figure contains three squares with area of 100, 16 and 49 sq. units respectively laying side by side as shown. By how much should the area of the middle square be reduced in order that the total length PQ of the resulting three square is 19?



(a) 12 (b) 4
(c) 2 (d) $\sqrt{2}$

Ans. (a): Let the final length of the side of the smaller square be a
Now, a + 10 + 7 = 19
A = 19 – 17 = 2
∴ Area of the smaller square = $(2)^2 = 4$
∴ Decrease in the area of the smaller square = 16 – 4 = 12 units.

138.The average of x_1, x_2 and x_3 is 14. Twice the sum of x_2 and x_3 is 30. What is the value of x_1.

(a) 20 (b) 27

(c) 16 (d) 2

Ans. (b): $\frac{x_1 + x_2 + x_3}{3} = 14$
$x_1 + x_2 + x_3 = 42$.....(i)
$x_2 + x_3 = 42 - x_1$.....(ii)
$2(x_2 + x_3) = 30$
$x_2 + x_3 = 15$
then, $(x_1 + x_2 + x_3 - x_2 + x_3) = 42 - 15 = 27$

139.A rectangle has perimeter of 50 meters. If its length is 13 meteres more than its breadth, then its area is:

(a) 124 m^2 (b) 144 m^2
(c) 114 m^2 (d) 104 m^2

Ans. (c): Let the breath be b.
Then, length = b + 13
Perimeter = 50 = 2(l + b)
2(b + 13 + b) = 50
L = 6 + 13 = 19m
Area = length × breath
= 19 × 6 = 114 m^2

140.Two packs of cards are thoroughly mixed and shuffled and two cards are drawn at random, one after the other. What is the probability that both of them are jacks?

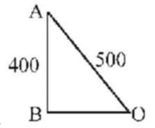
(a) 1/13 (b) 2/13
(c) 7/1339 (d) 1/169

Ans. (c): Total number of cards = 104 = 2×52
and total number of jacks = 8 = 2×4
∴ probability for the jack in first Draw = 8/104
and probability for the jack in second draw = 7/103
Since both the events are independent events.
Hence the probability that both of them are jacks
$\frac{-8}{104} \times \frac{7}{103} = \frac{7}{1339}$.

141.A man starts walking in the north easterly direction from a particular point. After waling a distance of 500 meters, he turns South Wards and walks a distance of 400 meters. At the end of this walk, he is situated:

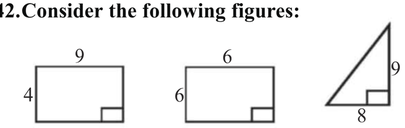
(a) 300 meters north of the starting point.
(b) 100 meters north-east of the starting point.
(c) 300 meters east of the starting point
(d) 100 meters north of the starting point

Ans. (c): According to pythagoras theorem



$OA^2 = AB^2 + OB^2$
$(500)^2 = (400)^2 + (OB)^2$
$OB^2 = 25000 - 16000$
$OB^2 = 9000$
OB = 300 meter

142.Consider the following figures:



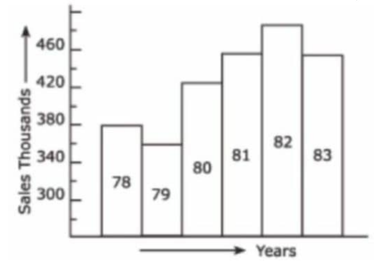
(i) (ii) (iii)

Which one of the following conclusions can be drawn from these figures?

(a) The area of the three figure are all different
(b) The area of all three figure are equal
(c) The perimeter of the three figures are all equal
(d) The perimeters of figures I and II are equal.

Ans. (b): Area of rectangle = 9 × 4 = 36
Area of square = 6 × 6 = 36
Area of triangle = ½ × 9 × 8 = 36
∴ Area of all three figure are equal.

143.The following figure represents sales (is thousands). Over the period 1978 to 1983. The sales in 1981 exceeded that in 1979 by



(a) Rs. One hundred (b) Rs. Then thousand
(c) Rs. One lakh (d) Rs. Ten lakh

Ans. (c): Sales in year 1979 = 320,000
Sales in year 1981 = 420,000
Thus the required difference
= 420,000 – 320,000
= Rs. 100,000

144.In an accurate clock, in a period of 2 hours and 20 minutes (140 minutes), the minutes hand will move over.

(a) 520^0 (b) 320^0
(c) 840^0 (d) 140^0

Ans. (c): The minute hand will move over in